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Capacity optimization of a wind-photovoltaic-electrolysis-battery (WPEB) hybrid energy system for power and hydrogen generation

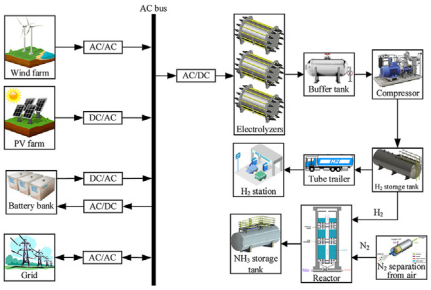
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HIGHLIGHTS

- WPEB system capacity optimization method is proposed.
- The wind-solar complementary metric is used to form a typical meteorological year.
- Increased wind power capacity reduces LCOE, grid sales capacity and net generation volatility.
- The optimal case is W190—P10-E95-B30 with LCOE and LCOH of 0.2692¥/kWh, 14.1574¥/kg.
- The IRR and ROI of W190—P10-E95-B30 are 12.60% and 78.97%.

GRAPHICAL ABSTRACT



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ABSTRACT

Utility-scale (>10 MW) Wind-Photovoltaic-Electrolysis-Battery (WPEB) system is an emerging technology that adopts open loop "Power-to-H_2" architecture for large-scale green hydrogen production. It applies to curtailment reduction in the area with abundant wind and solar energy resources. The traditional residential-scale (0—1 MW) or commercial/facility-scale (1—10 MW) WPEB systems usually adopt the close loop "Power-to-H_2-to-Power" structure for power supply in the microgrid. The capacity optimization of the commercial-scale, residential-scale WPEB system have been studied extensively, but

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Minimize levelized cost of electricity (LCOE)
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the researches for utility-scale WPEB system are rare. This work proposes a multi-constraints single objective capacity optimization method for the utility-scale WPEB system. The optimization objective is to minimize the levelized cost of electricity (LCOE) and the constraints of power balance, annual grid sales rate (AGSR), wind/PV/electrolysis station capacity are considered. The feasible domain is identified by the 8760-h production simulation and the gradient descent method is applied to search the optimal solution. The main conclusions are summarized below: (1) Increasing the wind-to-generation capacity ratio can reduce the LCOE, required grid sales capacity, and net generation curve fluctuation. (2) The lower and upper limits of the electrolysis station capacity are determined by the AGSR constraint and the LCOE target. (3) The battery bank time-shifts electricity to reduce the downtime of the electrolysis station. Thus, its power capacity should be no less than the minimum load ratio of the electrolysis station and the discharge duration greater than 1 h is preferred. Overall, this work provides a capacity optimization roadmap for the utility-scale WPEB system which plays a significant role in renewable electricity-based hydrogen production and curtailment reduction.

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Nomenclature

Symbols

$Capacity_{wind}$	the capacity of the wind farm
$Capacity_{PV}$	the capacities of the PV farm
$Capacity_{electrolyzer}$	the capacities of the electrolysis station
$C_{annualized,nominal}$	the annualized nominal cost of the system
$E_{grid,sales}$	electricity sales to the grid
E_{PV}	electrical production from PV farm
E_{served}	the total electric load served
$E_{served,ACprim}$	AC primary load served
$E_{served,DCprim}$	DC primary load served
$E_{served,def}$	deferrable load served
E_{wind}	electrical production from wind farm
$f_{global,n}$	the occurrence probability of the nth category Pearson coefficient during 2001–2021
$f_{t,1}, f_{t,2}, f_{t,n}$	the occurrence probability of the nine Pearson coefficient classes
n	the nth category Pearson coefficient classes
$P_{battery\ loss}$	inverter loss
$P_{electrolyzer\ load}$	electrolyzer load
$P_{grid\ purchase}$	power purchased from grid
$P_{grid\ sales}$	power sales to grid
$P_{plant\ load}$	battery power loss
P_{PV}	PV power output
$P_{rectifier\ loss}$	rectifier loss
P_{wind}	Wind power output
r_{xy}	Pearson coefficient
x	the wind speed at a specific height above the ground
$\overline{x}$	the mean value of wind speed at a specific height above the ground
y	global horizon irradiance
$\overline{y}$	the mean value of global horizon irradiance

Abbreviations

AC	Alternating Current
AGSR	Annual Grid Sales Rate
BA	Bat Algorithm
CAPEX	Capital Expenditure
CF	Capacity Factor
COV	Coefficient Of Variation
CRF	Capital Recovery Factor
CRITIC	Criteria Importance Though Intercrieria Correlation
CS	Chaotic Search
DC	Direct Current
DE	Differential Evolution
ED	Euclidean distance
GA	Genetic Algorithm
GHI	Global Horizontal Irradiance
GHR	Global Horizontal Radiation
HCHSA	hybrid chaotic search/harmony search/simulated annealing
HVAC	High Voltage Alternating Current
HVDC	High Voltage Direct Current
IGDT	Information Gap Decision Theory
IRENA	International Renewable Energy Agency
BEIS	Department for Business, Energy and Industrial Strategy
IRR	Internal Rate of Return
LCOE	Levelized Cost of Electricity
LCOH	Levelized Cost of Hydrogen
LOLP	Loss Of Load Possibility
MOEA/D	Multi-Objective Evolutionary Algorithm based on Decomposition
MOPSO	Multi-Objective Particle Swarm Optimization
NASA	National Aeronautics and Space Administration
NPC	Net Present Cost
NPV	Net Present Value
NSGA-II	Non-Dominated Sorting Genetic Algorithm II
O&M	Operating & Maintenance
POWER	Prediction of Worldwide Energy Resources
PSO	Particle Swarm Optimization
PV	Photovoltaics

ROI	Return On Investment	TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
SOC	State Of Charge		
SPEA-II	Strength Pareto evolutionary algorithm	WPEB	Wind-Photovoltaic-Electrolysis-Battery
STDEV	Standard Deviation		

1. Introduction

China is striving to reach peak CO_2 emission before 2030 and achieve carbon neutrality before 2060 [1–3]. China has accelerated the energy scheme transition from a fossil fuel-dominant system to a renewable energy-dominant system since 2020. The wind & solar energy installed capacity in 2020 (planning base year) is 535.21 GW and the planned installed capacity in 2030 is 1200 GW [4]. The average annual increment is 66.479 GW as shown in Fig. 1. China has planned two batches of large-scale wind & photovoltaic (PV) power bases in 2021 and 2022 respectively, each batch has a total installed capacity of 97.05 GW and 165 GW [5,6]. These large-scale wind & PV power bases concentrate on desertified land, Gobi desert, desertification land, coal mining subsidence area. The desertified land and desertification land cover 1.69 million km^2 and 2.57 million km^2 which account for 17.58% and 26.81% of China's land area [7]. The vast majority of the large-scale wind & PV power bases are located in the northern provinces (e.g. Inner Mongolia Autonomous Region, Shanxi) and northwestern provinces (e.g. Ningxia, Shaanxi, Gansu, Qinghai, Xinjiang) [8,9] due to the premium wind and solar energy resource as shown in Fig. 2 (a). However, the bases are far away from the load centers in the Southeast coastal areas as shown in Fig. 2 (b) [10,11]. The inverse distribution between wind-solar resource endowment and electric load centers leads to a huge wind and solar energy curtailment in the areas with abundant wind and solar resources. Inner Mongolia Autonomous Region (including two balancing regions of Mengxi and Mengdong) has the maximum wind energy curtailment of 6.09 TWh (5.06 TWh for Mengxi, 1.03 TWh for Mengdong) in 2021 and their wind energy curtailment rates are 8.9% and 2.4% as shown in Fig. 3 (a). The solar energy curtailment is 0.59 TWh (0.56 TWh for Mengxi, 0.03 TWh for Mengdong) and the curtailment rate for Mengxi, Mengdong are 3.5%, 0.6% respectively.

So far Inner Mongolia launches 5 batches (the 1st batch issued in 2021, the 2nd, 3rd batches issued in 2022, the 4th, 5th batch issued in 2023) of wind-photovoltaic-electrolysis-battery (WPEB) system to reduce the curtailment rate [12–15]. The WPEB system utilizes wind & solar power to split water into hydrogen and oxygen. The total installed capacity of the 5 batches of WPEB projects is 13.6525 GW (wind: 8.9835 GW, PV: 4.669 GW) and the electrolyzers capacity accounts for 43.55% (5.945 GW) of the wind-PV capacity as shown in Table 1. The 5 batches of the WPEB system have the following characteristics:

1. Wind power makes up the majority of the total installed capacity due to the higher capacity factor. The wind power proportion is 16.65% of the 1st batch, but it grows

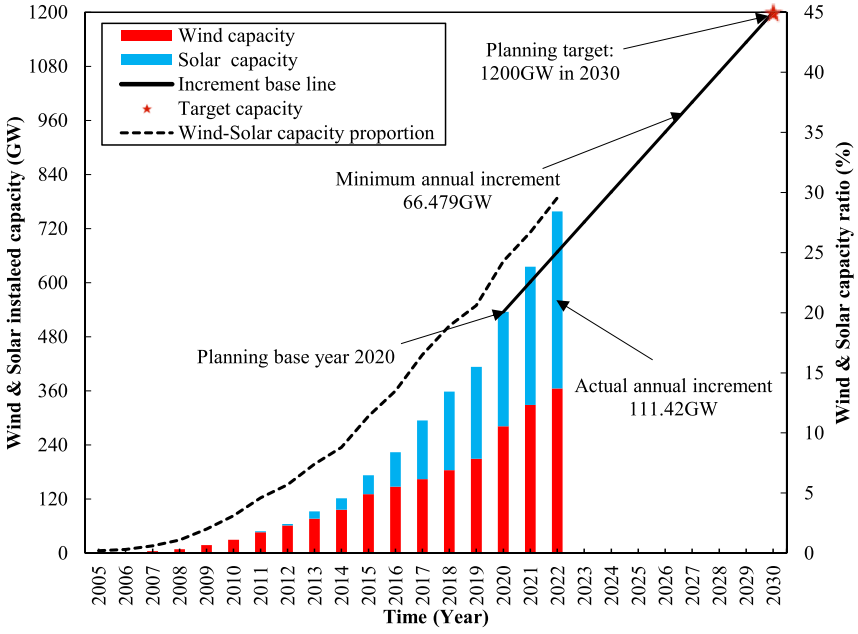


Fig. 1 – Comparison between planning (2020–2030) [4] and actual (2005–2022) [26–28] wind and solar installed capacity in China.

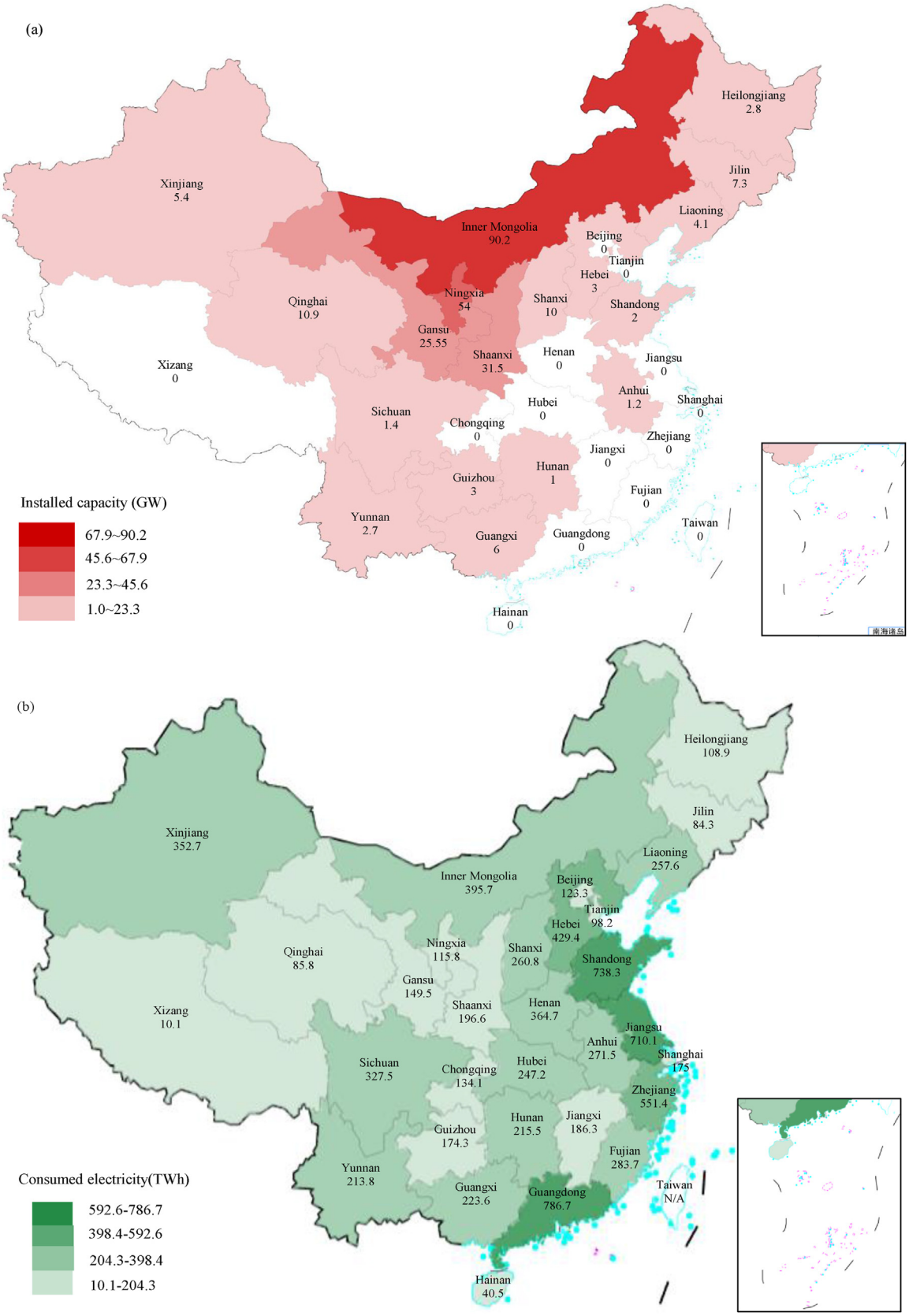


Fig. 2 – (a) Planned large-scale wind & PV power bases at the provincial level in China, data from [5,6], (b) provincial electricity consumption in 2021 [29].

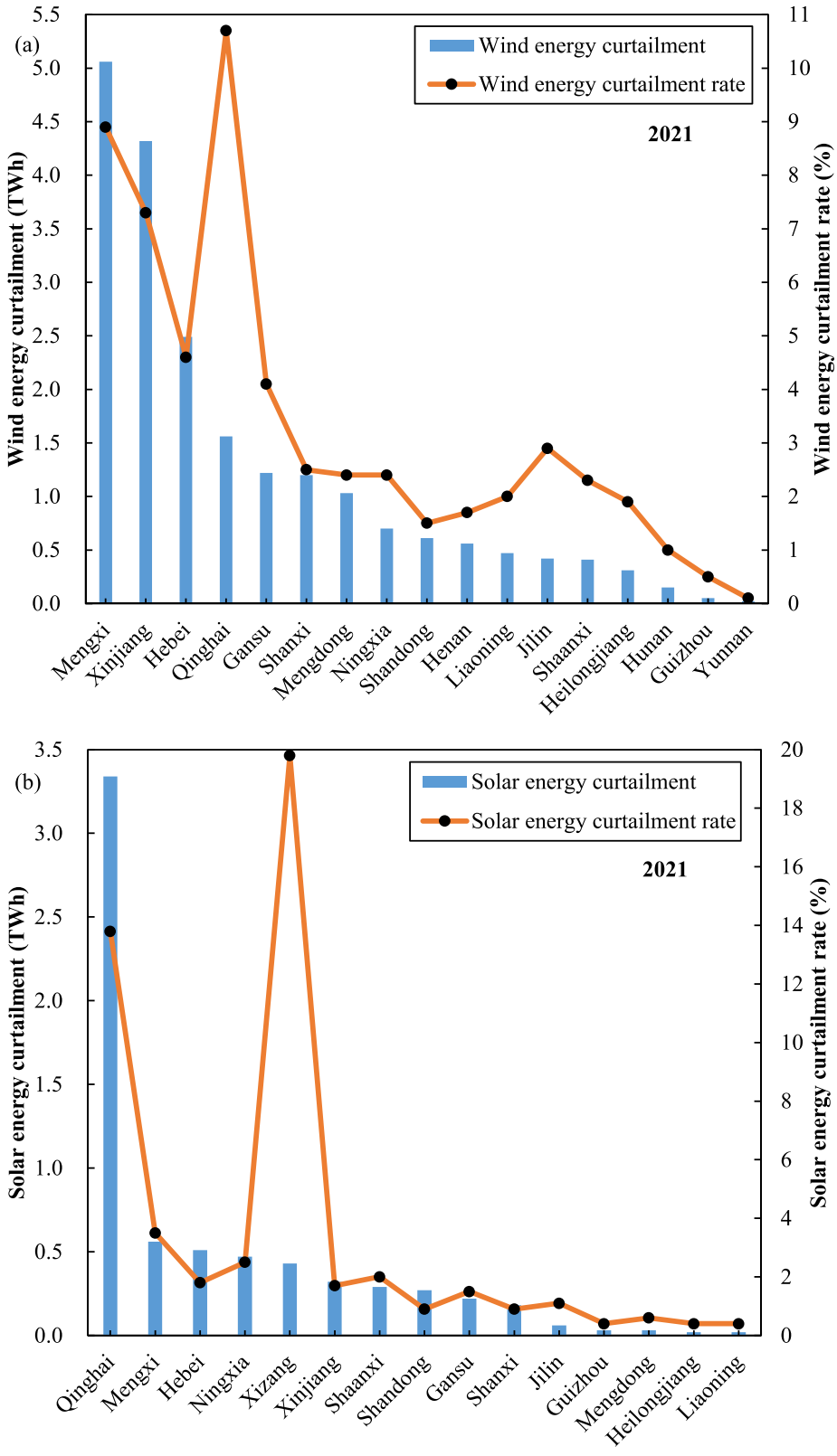


Fig. 3 – (a) Wind energy curtailment (rate); (b) solar energy curtailment (rate) at the provincial level in China (Inner Mongolia has two balancing regions of Mengxi and Mengdong), data from [29,30].

Table 1 – Overview of the recent (2021–2023) planned WPEB system in Inner Mongolia Autonomous Region in China.

No.	Type	Location	Wind (MW)	PV (MW)	Electrolysis (MW)	Battery bank (MW/MWh)	Stored H_2 (ton)	Produced H_2 (ton/year)	Ref
1.1	Grid-tied	Dalat Banner, Ordos	N/A	400	150	40/80	N/A	9300	[12]
1.2	Grid-tied	Wushen Banner, Ordos	49.5	270	160[a]	48/96[b]	N/A	10000	[12]
1.3	Grid-tied	Etok Front Banner, Ordos	N/A	250	95[a]	25/50[b]	N/A	6000	[12]
1.4	Grid-tied	Etok Front Banner, Ordos	N/A	250	95[a]	25/50[b]	N/A	6000	[12]
1.5	Grid-tied	Jungar Banner, Ordos	N/A	400	160[a]	40/80[b]	N/A	10000	[12]
1.6	Grid-tied	Damao Banner, Baotou	200	200	140	35/35	N/A	17800	[12]
1.7	Grid-tied	Damao Banner, Baotou	120	80	60	20/20	N/A	7800	[12]
2.1	Standalone	Huolin river, Tongliao	N/A	10	4[a]	1.5/6.0[b]	N/A	250	[13]
2.2	Standalone	Haggin banner, Ordos	N/A	100	45[a]	15/60[b]	N/A	2789.14	[13]
2.3	Grid-tied	Damao Banner, Baotou	500	N/A	224[a]	75/300[b]	N/A	22321	[13]
2.4	Grid-tied	Horqin Left Wing Middle Banner, Tongliao	140	60	90[a]	30/120[b]	N/A	8916.1	[13]
2.5	Grid-tied	Duolun County, Xilinguole League	100	N/A	35[a]	15/60[b]	N/A	3500	[13]
2.6	Grid-tied	Etok Front Banner, Ordos	500	5	200[a]	75.75/303[b]	N/A	20000	[13]
2.7	Grid-tied	Yijinhuoluo Banner, Ordos	N/A	260	83[a]	39/156[b]	N/A	5214	[13]
3.1	Grid-tied	Wushen Banner, Ordos	400	N/A	200[a]	60/240[b]	216000	20000	[14]
3.2	Grid-tied	Ulanqab	1742	804	1000[a]	381.9/1527.6[b]	N/A	100000	[14]
4.1	Standalone	Hinggan League	500	N/A	269[a]	100/400	8.99	26816	[15]
4.2	Standalone	Chagannaoer, Xilin gole League	12	N/A	4[a]	2/8	N/A	383.93	[15]
4.3	Standalone	Bayannur, Urad Middle Banner	400	100	360[a]	40/80	N/A	36000	[15]
4.4	Standalone	Bayannur, Urad Middle Banner	210	50	100[a]	39/39	N/A	10000	[15]
4.5	Grid-tied	Horqin Right Front Banner, Hinggan League	500	N/A	216[a]	50/100	24.72	21600	[15]
4.6	Grid-tied	Chifeng	450	50	242[a]	75/300	N/A	24200	[15]
4.7	Grid-tied	Chifeng	290	200	186[a]	73.5/73.5	N/A	18600	[15]
4.8	Grid-tied	Duolun County, Xilinguole League	120	30	55[a]	22.5/45	N/A	5419	[15]
4.9	Grid-tied	Ejin Horo Banner, Ordos	125	50	55[a]	26.25/105	N/A	5445	[15]
4.10	Grid-tied	Ordos	225	400	332[a]	2.5/5.0	19.77	21000	[15]
4.11	Grid-tied	Baotou	500	N/A	280[a]	75/150	2.7	28009	[15]
4.12	Grid-tied	Xinghe County, Ulanqab	350	150	257[a]	50/200	N/A	25700	[15]
4.13	Grid-tied	Alashan	400	200	223[a]	60/120	17.98	22300	[15]
4.14	Grid-tied	Tengger, Alashan	400	200	150	60/120	21.432	20827	[15]
4.15	Grid-tied	Kubuqi, Ordos	250	150	205	40/40	26.02	15460	[15]
5.1	Grid-tied	Hinggan League	500	0	270	75/300	N/A	22100	[31]
Total	N/A	N/A	8983.5	4669	5945	1716.9/5269.1	216121.612	553750.17	N/A

[a] The electrolysis station capacity is undisclosed and the value is estimated based on the required electricity for hydrogen production. The default electrolyzer efficiency is 75%. The electrolyzer capacity factors are 38% and 60% for the PV capacity dominant system and wind capacity dominant system.

[b] The battery bank power and energy capacities are undisclosed and the values are estimated based on the requirement of the regulations issued by the Energy Bureau of Inner Mongolia Autonomous Region.

massively to 74.03%, 72.71%, 74.97%, 100% in the 2nd, 3rd, 4th, 5th batches.
2. The power and energy capacity of the battery bank increase along with higher renewable energy penetration. The power capacity and discharge duration of the battery bank is 10% of the wind-PV power installed capacity and 1 h of the 1st batch, they further increase to 15% and 4 h of the rest of the batches.
3. The electrolysis station only operates on renewable electricity and the excess electricity is sold to the grid. The 1st batch of projects requires that the annual grid sales rate (AGSR) to the grid should be less than 20% of the generated electricity and there is no constraint on the purchased electricity from the grid. But the 2nd, 3rd 4th, 5th batches further demand that the WPEB system cannot purchase electricity from the grid except in emergency conditions.
4. The utility-scale WPEB system mainly applies to hydrogen generation and curtailment reduction in the area with abundant wind and solar energy resources, power supply to the grid is not the primary mission. The utility-scale WPEB system is developed to consume renewable electricity which is the key difference between the utility-scale WPEB in this work and the residential/commercial/facility-scale WPEB system. It is preferred that the utility-scale WPEB system can operate at the off-grid mode and reduce the peak regulation pressure of the bulk electric grid.

The WPEB system can be divided into 3 categories by size as residential-scale (0–0.1 MW), commercial/facility-scale (0.1–10 MW), and utility-scale (≥10 MW) [16]. Capacity configuration plays a key role in the WPEB system planning & design which influences the project's economic viability. The published articles on WPEB system capacity optimization are summarized in Table 2. There are four major differences between the published papers and this work:

1. Application scenario: The majority of the WPEB systems (such as case 1-case 20 in Table 2) are designed for reliable power supply to islands, buildings, communities, business parks, industrial parks, etc Their role is similar to the traditional hybrid energy system for power generation only [17,18]. The hydrogen only acts as a long-term energy carrier and then converted back to electricity. This work uses WPEB system to produce large-scale hydrogen for curtailment reduction of the bulk power grid and the hydrogen is used for fuel and chemical feedstock in the transportation and industrial sectors.
2. System architecture: The WPEB systems for power supply usually adopt a close loop "Power-to-H_2-Power" structure which employ fuel cell or hydrogen engines to convert the produced hydrogen back to power. This work applies the open loop "Power-to-H_2" structure which electrolyzers operate on renewable electricity for hydrogen production. The produced hydrogen is mainly applied as chemical feedstock for methanol, ammonia and other electro-fuels (hydrocarbons, alcohols, biodiesels, etc.) synthesis [19–21]. The surplus electricity is sold to the grid.
3. System size: The majority of the capacity optimization research focus on the residential-scale (0–0.1 MW), commercial/facility-scale (0.1–10 MW) WPEB systems such as case 1-case 19, case 22-case 23 in Table 1. This work applies to the utility-scale (≥10 MW) WPEB systems which utilize large-scale renewable energy for hydrogen production and reduce the curtailment of the bulk power grid.
4. Optimization method: The optimization methods adopted in the previous work include techno-economic analysis, genetic algorithms, evolutionary algorithms (NSGA-II, Differential Evolution, SPEA-II), heuristic algorithm (PSO, Bat Algorithm), simulated annealing algorithms, weighting methods (CRITIC), decision analysis methods (TOPSIS, IGDT), hybrid optimization methods (HCHSA). This work combines the 8760-h production simulation and gradient descent method. The optimized capacity is based on the long-term (8760 h) wind & solar energy resource rather than the typical day or week and the volatility, intermittency, randomness of renewable energy are considered [22].

WPEB hybrid energy system is an emerging technology for sustainable power and hydrogen cogeneration. But three knowledge gaps need to be addressed: (1) The capacity configuration and source-grid interaction mechanisms of the utility-scale grid-tied WPEB systems for large-scale hydrogen production [23]. (2) The combination and roles between short-term energy storage of battery and long-term energy storage of hydrogen [24]. (3) WPEB system is technically feasible but the cost reduction pathway needs to be proposed to make it economic competitiveness [25].

This work proposes a multi-strained single-objective capacity optimization method for a utility-scale grid-tied WPEB system. The objective is to minimize the LOCE and the decision variables are the wind-to-generation capacity ratio and electrolyzer-to-generation capacity ratio. The constraints mainly include power balance constraint, annual grid sales rate constraint, total installed capacity constraint, electrolysis capacity ratio constraint, etc. The research framework is outlined below: Section 1 briefly introduces the renewable energy curtailment condition in the northern and northwestern provinces of China. The utility-scale WPEB systems deployed in Inner Mongolia for hydrogen production and the capacity optimization methods are reviewed. Section 2 presents the WPEB system configuration, components & specifications (section 2.1), wind/solar resources analysis (section 2.2), and the multi-constraints single objective optimization method (section 2.3). Section 3 elaborates on the capacity optimization process (section 3.1), the impact of the WPEB system on renewable electricity grid integration and the cost reduction pathway (section 3.2), the limitation of this study and the prospect (section 3.3). Section 4 provides the concluding remarks.

Table 2 – Summary on the capacity optimization of the WPEB system.

No.	Structure	Scale	Application scenario	Objective function	System component	Optimization method	Installed capacity (kW)	Status	Ref.
1	Power-to-H_2-to-Power	Residential	Power supply	NPC, LCOE, power shortage rate, curtailment rate	WT, PV, electrolysis station, hydrogen tank, FC, battery	Analytic algorithm	2.2 (0.7 + 1.5)[a]	Off-grid	[32]
2	Power-to-H_2-to-Power	Residential	Power supply	NPC	WT, PV, electrolysis station, hydrogen tank, FC, diesel generator, battery	Techno-economic analysis	6 (3 + 3)	Off-grid	[33]
3	Power-to-H_2-to-Power	Residential	Power supply	NPC	WT, PV, electrolysis station, hydrogen tank, FC, battery	HCHSA	8.12 (8 + 0.12)	Off-grid	[34]
4	Power-to-H_2-to-Power	Residential	Power supply	LCOE	WT, PV, electrolysis station, hydrogen tank, FC, battery	Techno-economic analysis	10.24 (0 + 10.24)	Off-grid	[35]
5	Power-to-H_2-to-Power	Residential	Power supply	LCOE	WT, PV, electrolysis station, hydrogen tank, FC, battery, diesel generator	Techno-economic analysis	13 (3 + 10)	Off-grid	[36]
6	Power-to-H_2-to-Power	Residential	Power supply	NPC, LCOE	WT, PV, electrolysis station, hydrogen tank, hydrogen generator, battery	Techno-economic analysis	20 (20 + 0)	Off-grid	[37]
7	Power-to-H_2-to-Power	Residential	Power supply	NPC, LOLP	WT, PV, electrolysis station, hydrogen tank, FC, battery, power grid	Genetic algorithm	70 (46 + 24)	Grid tied	[38]
8	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC, LCOE	WT, PV, electrolysis station, hydrogen tank, FC, diesel generator	Techno-economic analysis	123 (120 + 3)	Off-grid	[39]
9	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC	WT, PV, electrolysis station, hydrogen tank, FC, battery,	Techno-economic analysis	273 (200 + 73)	Off-grid	[40]
10	Power-to-H_2-to-Power	Commercial/Facility	Power supply	LCOE	WT, PV, electrolysis station, hydrogen tank, FC	Techno-economic analysis	316.7 (300 + 16.7)	Off-grid	[41]
11	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC, curtailment rate, self-balance degree	WT, PV, electrolysis station, hydrogen tank, FC, battery, power grid	NSGA-II, MOPSO	About 400	Grid tied, off-grid	[42]
12	Power-to-H_2-to-Power	Commercial/Facility	Power supply	LCOE, LOLP, curtailment rate	WT, PV, electrolysis station, hydrogen tank, FC	NSGA-II, CRITIC, TOPSIS	443.525 (360 + 83.525)	Off-grid	[43]
13	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC, renewable energy consumption rate, LOLP	WT, PV, electrolysis station, hydrogen tank, FC, battery	NSGA-II, PSO	About 450	Off-grid	[44]
14	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC, LCOE	WT, PV, electrolysis station, hydrogen tank, FC, battery	Techno-economic analysis	960 (660 + 300)	Off-grid	[45]
15	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC, self-balance degree	WT, PV, electrolysis station, hydrogen tank, FC, battery	NSGA-II, MOSSA	About 1300	Off-grid	[46]
16	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC, LCOE, LCOH	WT, PV, electrolysis station, hydrogen tank, FC	Techno-economic analysis	1524.28(924.28 + 600)	Off-grid	[47]
17	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC	WT, PV, electrolysis station, hydrogen tank, FC, battery, power grid	PSO	1670 (900 + 770)	Grid tied	[48]

18	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC, power shortage rate, curtailment rate	WT, PV, electrolysis station, hydrogen tank, FC, battery, power grid	PSO, GA, DE, BA	4112.17 (2512.52 + 1599.65)	Grid tied	[49]
19	Power-to-H_2-to-Power	Commercial/Facility	Power supply	NPC	WT, PV, electrolysis station, hydrogen tank, FC	CS-PSO, NSGA-II	5,938 (3307 + 2631)	Off-grid	[50]
20	Power-to-H_2-to-Power	Utility	Power supply	LCOE, loss of power supply probability	WT, PV, electrolysis station, hydrogen tank, FC	NSGA-II, MOEA/D, MOPSO, SPEA-II	639360 (349540 + 289820)	Off-grid	[51]
21	Power-to-H_2	N/A	H_2 production	NPC, load loss rate, power shortage rate, curtailment rate	WT, PV, electrolysis station, battery, power grid	N/A	N/A	N/A	[52]
22	Power-to-H_2	Residential	H_2 production	NPC, LCOE, LCOH	WT, PV, electrolysis station, reformer, hydrogen tank, battery, power grid	Techno-economic analysis	36 (34 + 2)	Grid tied	[53]
23	Power-to-H_2	Commercial/Facility	H_2 production	LCOH	WT, PV, electrolysis station, battery	Techno-economic analysis	1704 (800 + 904)	Off-grid	[54]
24	Power-to-H_2	Commercial/Facility	H_2 production	NPC, LCOE, LCOH	WT, PV, electrolysis station, hydrogen tank, battery	Techno-economic analysis	1871 (1600 + 271)	Off-grid	[55]
25	Power-to-H_2	N/A	H_2 production	LCOE	WT, PV, electrolysis station, hydrogen tank, battery	Techno-economic analysis	N/A	Off-grid	[56]
26	Power-to-H_2	Utility scale	H_2 production	NPC	Renewable power source, electrolysis station, hydrogen tank	CPLEX	About 250,000	Off-grid	[57]
27	Power-to-H_2	Utility scale	H_2 production	NPV	WT, PV, electrolysis station, hydrogen tank, power grid	IGDT	403,000 (347000 + 56000)	Grid-tied	[58]
28	Power-to-H_2	Utility scale	H_2 production	LCOE	WT, PV, electrolysis station, hydrogen tank, power grid	Gradient descent & Techno-economic analysis	200,000	Grid-tied	This work

Abbreviation: HCHSA: hybrid chaotic search/harmony search/simulated annealing; NSGA-II: Non-Dominated Sorting Genetic Algorithm II; MOPSO: Multi-Objective Particle Swarm Optimization; CRITIC: Criteria Importance Though Intercrieria Correlation; TOPSIS: Technique for Order Preference by Similarity to Ideal Solution; PSO: Particle Swarm Optimization; GA: Genetic Algorithm, DE: Differential Evolution, BA: Bat Algorithm; CS: Chaotic Search; IGDT: Information Gap Decision Theory, MOEA/D: Multi-objective evolutionary algorithm based on Decomposition, SPEA-II: Strength Pareto evolutionary algorithm.

[a] A (B + C): "B" and "C" denote the wind and solar power installed capacity and the "A" is the total installed capacity.

2. Methodology

2.1. Study area, system configuration and specification

The studied site locates in the Damao Banner, Baotou City, Inner Mongolia Autonomous Region, China. The practical WPEB system has 120 MW wind farm, 80 MW PV farm, 60 MW electrolysis station, 20 MW battery bank and the techno-economic performance has been explored in ref. [59]. This work maintains the total wind-PV installed capacity constant of 200 MW and the capacities of wind power, PV, electrolysis station and battery bank are resized to minimize LCOE. The schematic diagram of the WPEB system is shown in Fig. 4. The wind farm and PV farm connect to the AC bus through the AC to AC converter and the DC to AC inverter respectively. The electrolysis station composes a series of 5 MW electrolyzers (the number depends on the electrolysis station capacity) which consume electricity to split the water into hydrogen and oxygen. The hydrogen flow rate is controlled by the buffer tank and the hydrogen is transported to the hydrogen storage system. The hydrogen is compressed to 80 MPa and stored in a high-pressure tank. There are two major hydrogen loads in the WPEB system, one is the hydrogen fueling station in the transportation sector and the other is the ammonia synthesis in the industry sector. The components specifications of the WPEB system are shown in Table 3. The electrolyzer efficiency (the ratio of the higher heating value of the produced hydrogen to the consumed electricity) is 75% and the cost by type of system components is provided in Table 4. The nominal discount rate, expected inflation rate and real discount rate are 8%, 3.5%, 4.35%. The project's lifetime is 25 years.

2.2. Resource Assessment

The summer of the Damao Banner ranges from June to August while the winter ranges from December to February [60]. Wind speed at 50 m and global horizontal irradiance (GHI) are selected as the representative metrics of wind and solar energy resources. Hourly data from 2001 to 2021 are collected from National Aeronautics and Space Administration (NASA) Prediction of Worldwide Energy Resources (POWER) Project [61].

2.2.1. Wind energy resource

The mean wind speed and mean wind power density are 7.56 m/s and 353.81W/m^2 of the studied site. The wind energy resource is classified as "Class 4 Good" according to the national standard GB/T 18710–2002 [62] which is suitable for the vast majority of wind turbine applications. The wind speed and wind power density indicate significant seasonal variation across the year. The windiest season ranges from March to May with an average wind speed of 8.22 m/s, 8.47 m/s, 8.46 m/s as shown in Fig. 5 (a). November and December are the windier months with an average wind speed of 8.23 m/s and 8.19 m/s. All these five months have a bulge from 9:00 to 16:00 as shown in Fig. 5 (b) which coincidences with the solar PV power output. The capacities of the wind farm and PV farm need to be designed carefully to reduce the daily power bulge at 9:00–16:00. July and August in is the calmest months with average wind speeds of 6.05 m/s, 6.08 m/s. June and September are the calmer seasons with an average wind speed of 6.83 m/s, 6.85 m/s. These four months have troughs at the period of 7:00–17:00 which is complementary to the occurrence of solar radiation.

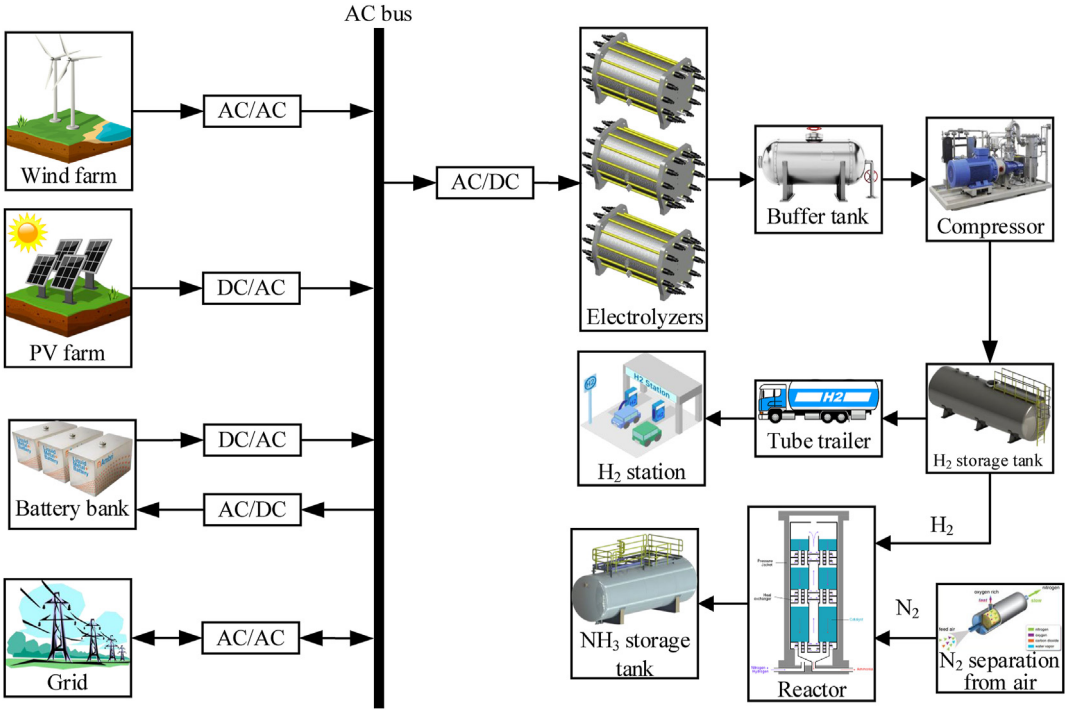


Fig. 4 – Schematic of wind-photovoltaic-electrolysis-battery (WPEB) hybrid energy system for power and hydrogen generation.

Table 3 – Components specifications of WPEB system.

Equipment	Parameter	Value
Wind turbine	Rated capacity (MW)	5
	Rotor diameter (m)	166
	Hub height (m)	105
	Rated wind speed (m/s)	10.5
	Cut-in wind speed (m/s)	2.5
	Cut-out wind speed (m/s)	25
	Extreme wind speed (m/s)	59.5
	Lifetime (year)	25
	No. of turbines	24
Flat plate PV arrays	Rated capacity on DC bus (MW)	96.15
	DC/AC ratio	1.2
	Rated capacity on AC bus (MW)	80
	Temperature coefficient of power (%/C)	−0.5
	Nominal operating cell temperature (C)	47
	Efficiency at standard test conditions (%)	13
	Derating factor (%)	90
	Ground reflectance (%)	20
	Tracking system	No tracking
	Lifetime (year)	25
System converter	Efficiency (%)	95
Electrolysis station	Rated capacity (MW)	60
	Efficiency (%)	75
	Minimum load ratio (%)	30
	Lifetime (year)	15
Battery cell	Nominal voltage (V)	51.2
	Nominal capacity (Ah)	240
	Roundtrip efficiency (%)	90
	Max charge current (A)	240
	Max discharge current (A)	240
	Lifetime (year)	10
Hydrogen tank	Capacity (kg)	1E+04
Bidirectional power converter	Efficiency (%)	95%

Table 4 – Cost by type of system components.

Components	CAPEX	Replacement cost	O&M	Lifetime (Years)
Wind farm	5917¥/kW	0	40.2¥/kW-year	25
Solar PV farm	4633¥/kW	0	17.6¥/kW-year	25
Electrolysis station	6964¥/kW	5969.14¥/kW	208.92¥/kW-year	15
Battery bank	2549¥/kW	500¥/kW	10¥/kW-year	10
Hydrogen storage system	6611.77¥/kg	0	141.68¥/kg-year	25

2.2.2. *Solar energy resource*

The annual global horizontal radiation (GHR) is 1731 Wh/m^2 (calculated from 2001 to 2021 data) which is classified as "Class B Very abundant" according to the national standard QX/T 89–2018 [63]. The brighter period lasts from April to July with daily radiation of 6.1638 kWh/m^2/day, 6.9201 kWh/m^2/day, 6.6419 kWh/m^2/day, 6.3643 kWh/m^2/day as shown in Fig. 6(a). November, December and January are the darker seasons with daily radiation of 2.7453 kWh/m^2/day, 2.1634 kWh/m^2/day, 2.4513 kWh/m^2/day. October has the most atmospheric clearness with a clearness index of 0.684 while the worst clearness period is July with a clearness index of 0.565. May, June and July have the longest PV power output period of 4:00–18:00 as shown in Fig. 6(b). The shortest PV power output period occurs in December of 6:00–15:00 and in February of 7:00–16:00.

2.2.3. *Typical meteorological year based on wind-solar complementarity*

Pearson coefficient r_{xy} is used as the wind and solar energy resource complementarity metric as shown in Eq. (1).

$$r_{xy} = \frac{\sum (x - \bar{x}) \cdot (y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \quad (1)$$

where x and y are the wind speed at a specific height above the ground and GHI while $\bar{x}$ and $\bar{y}$ are their mean values. Pearson coefficient r_{xy} ranges from −1 to 1 which is divided into 9 ranges as shown in Table 5. The value of −1 indicates a perfect negative linear relation that wind speed and GHI profiles are perfectly complementary. The value of 1 reflects a perfect positive linear relation between wind speed and GHI. The value of 0 represents no relation between the variables. The

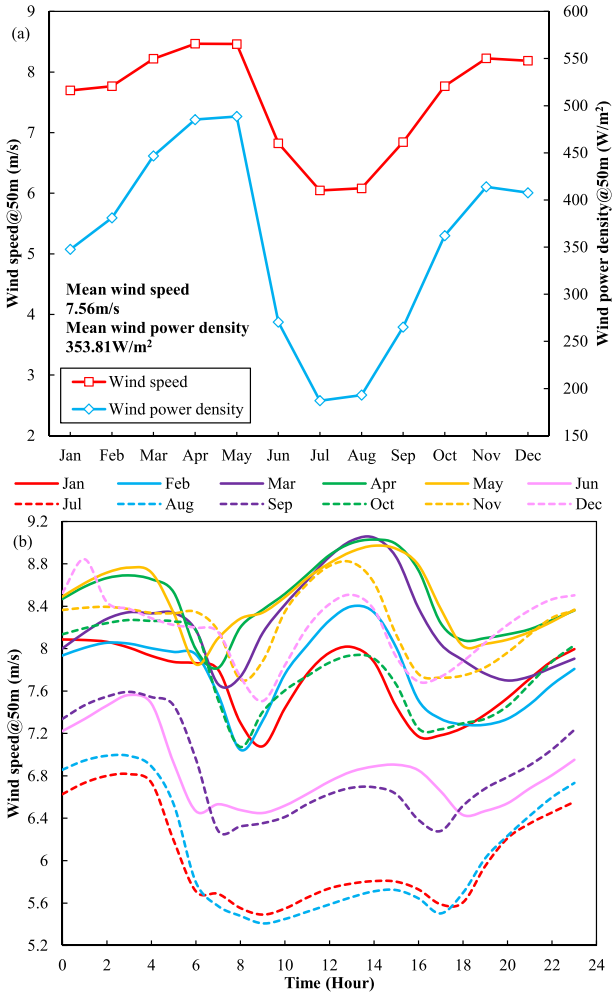


Fig. 5 – (a) Wind speed, wind power density at each month; (b) wind speed daily profiles (calculated from 2001 to 2021 data [61]).

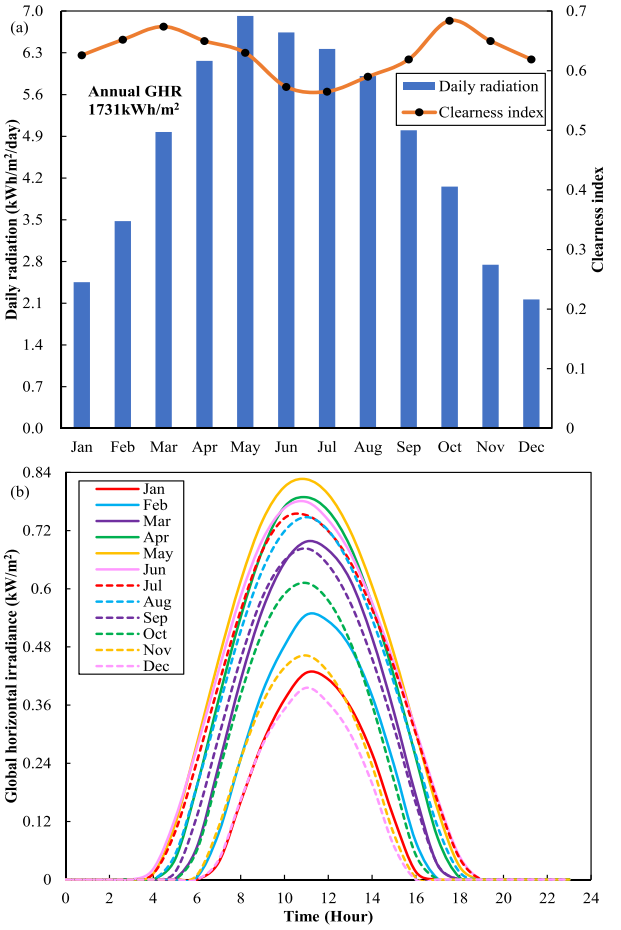


Fig. 6 – (a) Daily radiation and clearness index at each month; (b) solar irradiance daily profiles (calculated from 2001 to 2021 data [61]).

daily complementary metric during 2001–2021 and the long-term (2001–2021) average are calculated. Their probability of occurrence is presented in Fig. 7 (a). The weather pattern of the studied site shows an overall negative correlation between wind speed and GHI. The occurrence probability of $-1.0 \leq r_{xy} < -0.2$ and $0.2 \leq r_{xy} < 1.0$ are 41.47% and 36.73% respectively.

A real 8760-h wind speed and GHI data are used as model inputs to maintain the wind-solar complementary characteristic. Euclidean distance is used to measure the wind-solar complementarity between a particular year and the long-term average (2001–2021) as shown in Eq. (2). The wind speed and GHI data in 2008 are chosen as the typical meteorological year because its wind-solar complementarity is closest to the long-term trend as shown in Fig. 7 (b).

$$\text{Euclidean distance(ED)} = \sqrt{\sum_{n=1}^{9} \left(f_{t,n} - f_{global,n}\right)^2} \quad (2)$$

where $f_{t,1}, f_{t,2} \ldots f_{t,n} \ldots f_{t,9}$ are the occurrence probability of the 9 ranges of Pearson coefficient (see Table 5) of a particular year. n is the nth range of the Pearson coefficient and t is the tth year during 2001–2021. $f_{global,n}$ is the occurrence probability of the nth range of the Pearson coefficient during 2001–2021.

Table 5 – Classes of wind and solar energy resource complementarity.

Class	Pearson coefficient range	Correlation
1	$0.8 \leq r_{xy} < 1.0$	Very strong positive
2	$0.6 \leq r_{xy} < 0.8$	Strong positive
3	$0.4 \leq r_{xy} < 0.6$	Moderate positive
4	$0.2 \leq r_{xy} < 0.4$	Weak positive
5	$-0.2 \leq r_{xy} < 0.2$	Very weak or no correlation
6	$-0.4 \leq r_{xy} < -0.2$	Weak negative
7	$-0.6 \leq r_{xy} < 0.4$	Moderate negative
8	$-0.8 \leq r_{xy} < -0.6$	Strong negative
9	$-1.0 \leq r_{xy} < -0.8$	Very strong negative

2.3. *Multi-constraints single objective optimization*

The objective function is to minimize the levelized cost of electricity (LCOE) which is the ratio of the annualized nominal cost $C_{annualized,nominal}$ to the total electric load served E_{served} as shown in Eq. (3). The nominal annualized cost $C_{annualized,nominal}$ is the product of the capital recovery factor

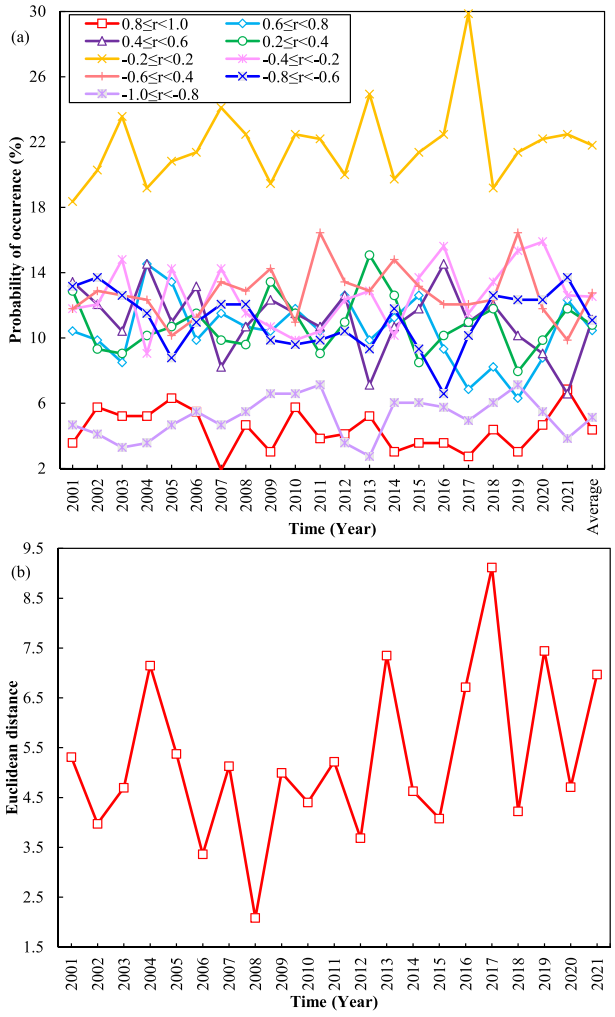


Fig. 7 – (a) Complementary metric of each year and long-term (2001–2021) average; (b) Euclidean distance between annual and long-term complementary.

(CRF) times the net present cost (NPC) as shown in Eq. (4). The total electric load served E_{served} includes AC primary load served $E_{served,ACprim}$, DC primary load served $E_{served,DCprim}$, deferrable load served $E_{served,def}$ and energy sold to the grid $E_{grid,sales}$ as shown in Eq. (5). The wind to generation capacity ratio and the electrolyzer to generation capacity ratio are the decision variables which are determined by the wind farm capacity and the electrolysis station capacity. Their initial search steps for wind farm and electrolysis station are 20 MW (5 MW wind turbine × 4) and 5 MW (stack sizes up to 5 MW in size) respectively and the former is dynamically adjusted to 10 MW (5 MW wind turbine × 2) to accelerate the solving process. The battery bank power capacity is set as 30% of the electrolysis station capacity which corresponds to the minimum load ratio of the electrolysis station. The electrolyzers scheduling strategy is not considered in this work. The battery bank plays the role of electricity time-shift to reduce the downtime of the electrolysis station as shown

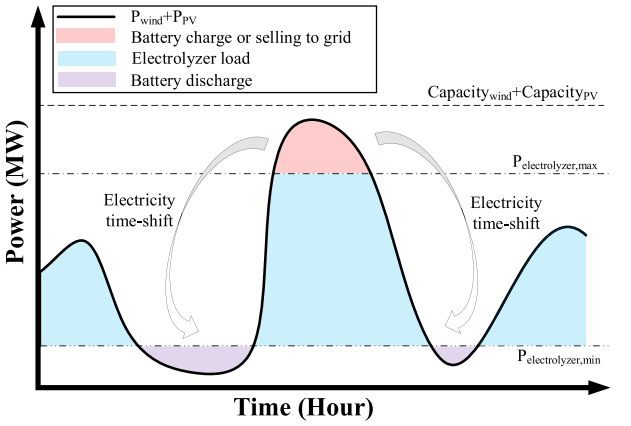


Fig. 8 – Exemplary sketch of electricity time shifting by battery bank to reduce the downtime of electrolyzers.

in Fig. 8. The load following dispatch strategy is adopted and the surplus electricity sales to the grid.

$$\min \textit{Levelized cost of electricity (LCOE)} = \frac{C_{annualized,nominal}}{E_{served}} \tag{3}$$

$$C_{annualized,nominal} = CRF(i,N) \times NPC \tag{4}$$

$$E_{served} = E_{served,ACprim} + E_{served,DC\ prim} + E_{served,def} + E_{grid,sales} \tag{5}$$

The WPEB system capacity sizing is restricted by the power balance, annual grid sales rate (AGSR), total installed capacity (wind + PV), electrolysis station capacity as shown in Eq. (6). The first formula describes the system power balance constraint. P_{wind}, P_{PV}, $P_{grid\ purchase}$, $P_{electrolyzer\ load}$, $P_{plant\ load}$, $P_{battery\ loss}$, $P_{inverter\ loss}$, $P_{rectifier\ loss}$, $P_{grid\ sales}$ are the wind power output, PV power output, power purchased from the grid, electrolyzer load, battery power loss, inverter loss, rectifier loss, power sales to the grid. The second formula describes that the annual grid sales rate should be no more than 20% which is required by the mandatory policy. $E_{grid\ sales}$, E_{wind}, E_{PV} are the electricity sales to the grid, electrical production from the wind farm, electrical production from the PV farm. The third formula is the total installed capacity constraint which sets as 200 MW in this work. $Capacity_{wind}$, $Capacity_{PV}$, $Capacity_{electrolyzer}$ are the capacities of the wind farm, PV farm, electrolysis station. The fourth formula is the electrolyzer-to-generation capacity ratio constraint which restricts the upper limit of the electrolysis station capacity [64].

The flow chart of the WPEB system capacity optimization is presented in Fig. 9. The electric or hydrogen load, system architecture, natural resource, economic parameters are the necessary input for model development. The WPEB system modeling methodology has been provided in ref. [59]. Then, the decision variables of wind to generation capacity ratio and the electrolyzer-to-generation capacity ratio are varied independently which form the solution set. The 8760-h system production simulation is performed to identify the feasible solutions which meet all the constraints in Eq. (6). The minimum levelized cost of electricity (LCOE) must appear at the

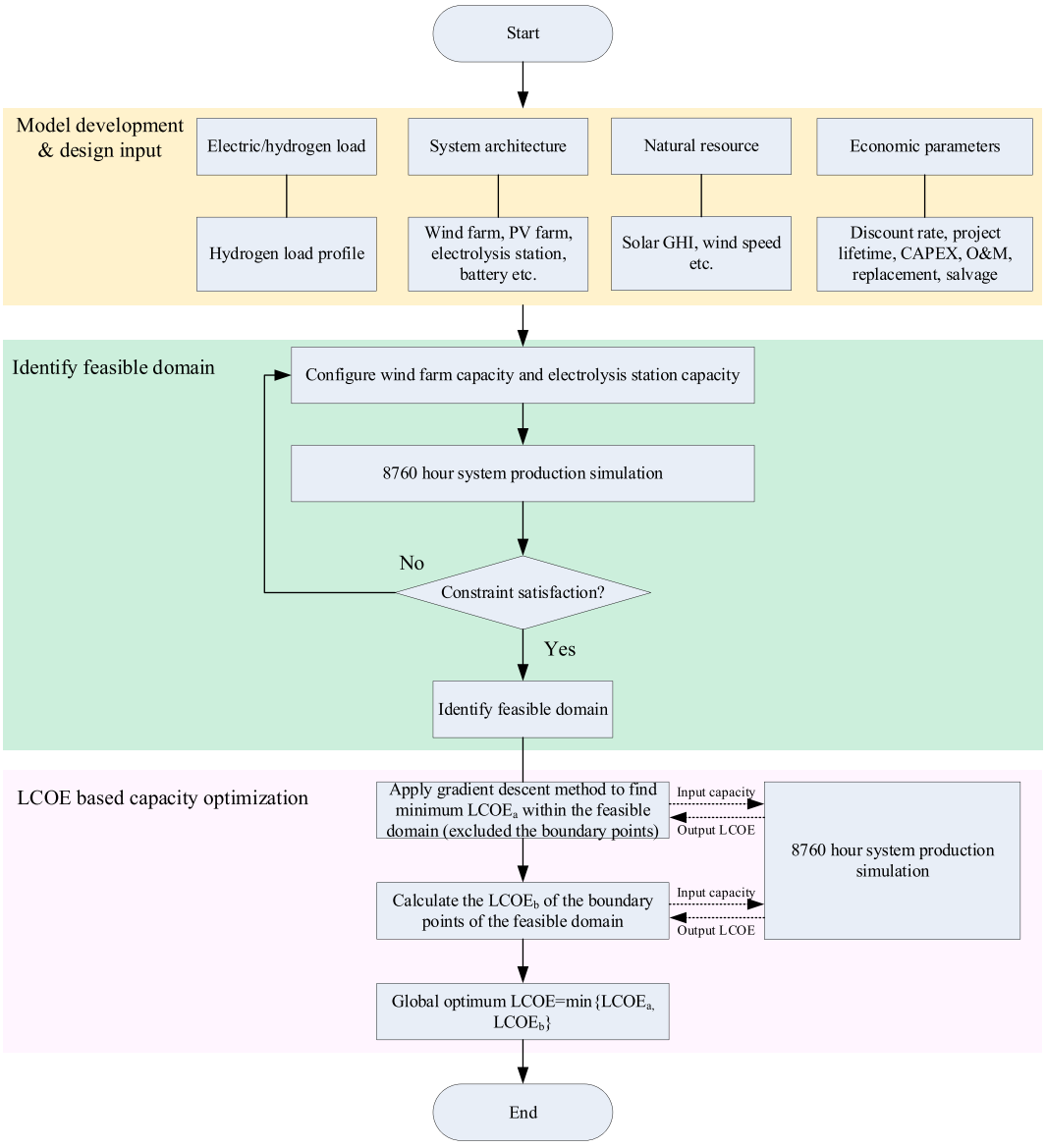


Fig. 9 – Flow chart of the WPEB system capacity optimization.

point with the lowest gradient or the boundary points. The minimum LCOE within the feasible domain is searched by the 8760-h production simulation and gradient descent method, then it will be compared with the LCOE of the boundary points. The case with the lowest LCOE is regarded as the global optimal solution.

$$\text{s.t.}\begin{cases} P_{wind}+P_{PV}+P_{grid\ purchase}=P_{electrolyzer\ load}+P_{plant\ load}+P_{battery\ loss} \\ \quad +P_{inverter\ loss}+P_{rectifier\ loss}+P_{grid\ sales} \\ Annual\ grid\ sales\ rate(\text{AGSR})=\dfrac{E_{grid\ sales}}{E_{wind}+E_{PV}}\times 100\%\le 20\% \\ Capacity_{wind}+Capacity_{PV}=200\text{MW} \\ 0<Capacity_{electrolyzer}\le 0.5\times\left(Capacity_{wind}+Capacity_{PV}\right) \\ Capacity_{wind}\ge 0 \\ Capacity_{PV}\ge 0 \end{cases} \tag{6}$$

3. Results and discussion

3.1. Main findings of the present study

The feasibility domain of the WPEB system is presented in Fig. 10. The electrolyzer-to-generation capacity ratio should be no less than 0.375 (corresponding to 75 MW electrolysis station) to address the requirement of annual grid sales rate below 20%. Correspondingly, the wind-to-generation capacity ratio is 0.5, namely 100 MW wind farm and 100 MW PV farm. The feasible wind-to-generation capacity ratio widens as the increasing electrolyzer to generation capacity ratio. For example, the feasible wind-to-generation capacity ratio ranges from 0.3 (corresponding to 60 MW wind farm) to 0.7 (corresponding to 140 MW wind farm) at the electrolyzer-to-generation capacity ratio of 0.425 (corresponding to 85 MW electrolysis station). The feasible wind-to-generation

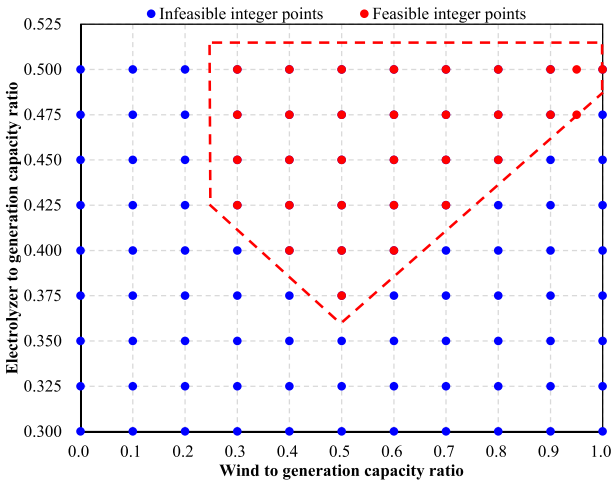


Fig. 10 – Search for the feasible integer points from variable space.

capacity ratio continues to widen when the electrolyzer-to-generation capacity ratio exceeds 0.425. But the feasible domain only expands on the wind farm side rather than the PV farm side. The PV power output occurs during the daytime and a greater electrolyzer capacity is required to reduce the electricity sales to the grid compared to wind power.

Increased wind-to-generation capacity ratio and reduced electrolysis station can decrease the LCOE within the feasible domain as shown in Fig. 11. The capacity factor (CF) of the wind farm and the PV farm are 40% and 19% and the higher CF favors LCOE reduction. The onshore wind and solar PV weighted average capacity factor in China, 2021 are 36.1% [65] and 16.02% [26]. The electrolysis station capacity should be minimized for LCOE reduction meanwhile satisfying the 20% annual grid sales rate constraint. The case of 190 MW wind farm-10 MW PV farm-95MW Electrolysis station-30MW/30 MWh battery bank (W190-PV10-E95-B30) obtains the minimum LCOE of 0.2692 ¥/kWh as shown in Fig. 11. The capacity factor of the electrolysis station is 62.40%. Reducing the electrolysis station capacity can increase its capacity factor, but the annual grid sales rate and transmission line capacity would increase. Overall, the electrolysis station capacity mainly depends on the application scenario and operational requirements. The electrolysis station operation has four scenarios: (1) Grid-connected scenario: electrolysis station is connected to the bulk electric grid. The electrolysis station operates at a stable power that has a CF greater than 90% [66,67]. The electrolysis station only operates during the off-peak period whose CF is around 40%–70% [66,67]. (2) Renewable energy connected scenario: electrolysis station is directly connected to renewable electricity and its capacity factor depends on renewable electricity generation. The electrolysis station CF ranges from 51% to 62% in this scenario [67]. (3) Curtailed electricity scenario: the electrolysis station only operates on curtailed electricity which has a low CF of about 8%–35% [66,67].

The cases with different wind-to-generation capacity ratios (W120-PV80-E80-B25, W60-PV140-E85-B27.5) are compared with the optimal cases to figure out the impact of wind/PV/electrolyzers capacities on the required grid sales capacity, total power output, net generation sales to grid, etc. The probability distributions of wind-PV power for three cases during 2001–2021 are shown in Fig. 12. The total electricity generation among 21 years of W190-PV10, W120-PV80, W60-PV140 are 13.72 TWh, 11.32 TWh, 9.26 TWh and their equivalent capacity factor are 37.29%, 30.77%, 25.18%. The dominant wind-PV power ranges of W190-PV10, W120-PV80, W60-PV140 are 120–130 MW, 70–80 MW, 30–40 MW as shown in Fig. 13. The maximum electric power generation of W190-PV10, W120-PV80, W60-PV140 are 154.07 MW, 176.93 MW, 199.91 MW and their required grid sales capacities are 59.07 MW, 96.93 MW, 144.91 MW respectively as shown in Fig. 12. The typical daily operation with a maximum wind-PV power output of W190-PV10-E95-B30, W120-PV80-E80-B25, W60-PV140-E85-B27.5 are present in Fig. 14 which use the weather pattern of 2021–01-06, 2021-01-06, 2012-02-06. The maximum net power output of W190-PV10-E95-B30, W120-PV80-E80-B25, W60-PV140-E85-B27.5 are 58.71 MW, 96.57 MW, 114.26 MW which is below the required grid sales capacity mentioned above. The average net generation, standard deviation (STDEV) and coefficient of variation (COV) of W190-PV10-E95-B30, W120-PV80-E80-B25, W60-PV140-E85-B27.5 are 15.37MW/14.60MW/0.9498, 9.29MW/16.77MW/1.8061, 10.64MW/18.65MW/1.7524 as shown in Fig. 15. Increasing the wind-to-generation capacity ratio can attenuate the net generation curve fluctuation. In addition, the battery bank discharge duration should be increased to reduce the downtime of the electrolysis station.

The general directions of capacity configuration forWPEB system are summarized below:

1. The optimization objective is to minimize the LCOE at the design planning stage which is mainly determined by the local wind/solar energy resource, performance parameters, technology-specific cost (CAPEX, O&M, Replacement cost, salvage), annual grid sales rate. However, the optimization objective is to maximize the NPV at the operation stage which is determined by the hydrogen selling price, feed-in tariff, tradable green certificate price, dispatching priority.
2. Increasing the wind-to-generation capacity ratio can take advantage of the high capacity factor of wind farm compared to PV farm.
3. Increasing the wind-to-generation capacity ratio can lower the required grid sales capacity and the resulting grid extension cost.
4. Increasing the wind-to-generation capacity ratio can reduce the net generation curve fluctuation which benefits a large-scale renewable energy integration.

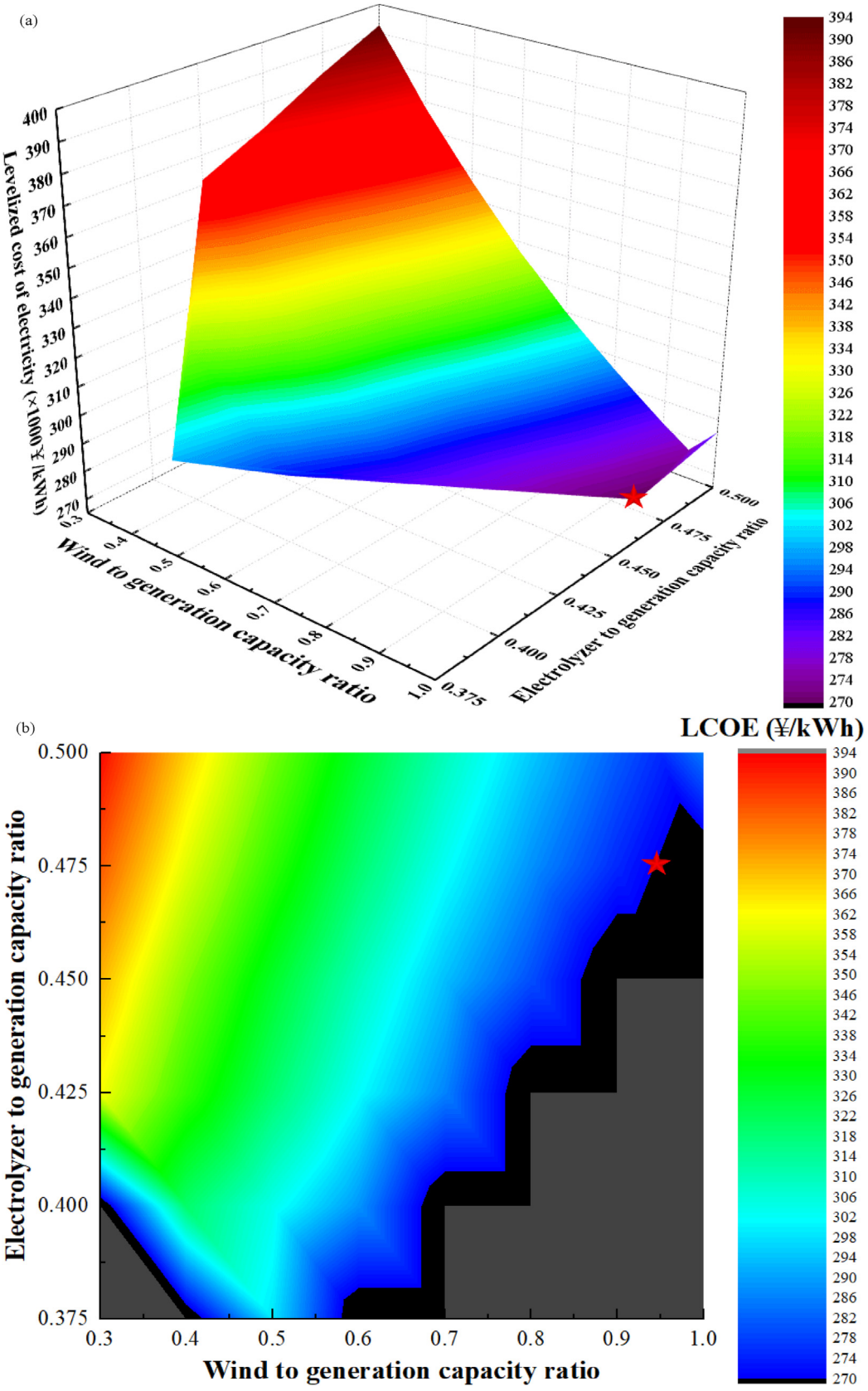


Fig. 11 – Impact of wind/PV/electrolyzer portfolio on LCOE (a) 3D map; (b) 2D contour map.

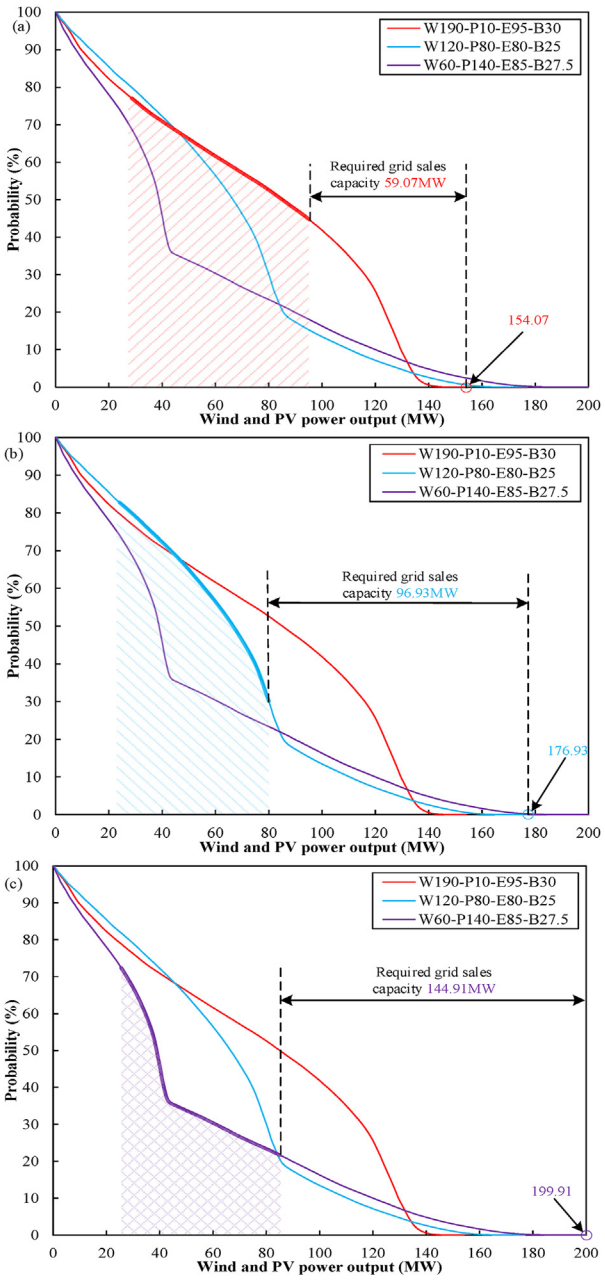


Fig. 12 – Probability distribution of wind-PV power generation using 2001–2021 weather patterns (the shaded area corresponding to the electrolysis station operating domain).

5. The electrolysis station capacity is co-determined by the annual grid sales rate constraint and the LCOE objective.
6. The battery bank power capacity should be no less than the minimum load ratio of the electrolysis station. The battery bank discharge duration should be no less than 1 h to reduce the downtime of electrolyzers.

3.2. *Comparison with other studies*

The economic performance of W190–P10-E95-B30 is superior to the W120–P80-E80-B25, W60–P140-E85-B27.5. The NPV of W190–P10-E95-B30, W120–P80-E80-B25, W60–P140-E85-B27.5 are 2113.08 million¥, 1605.46 million¥, 750.91 million¥ and their payback period are 9 years, 10 years, 13 years as shown in Fig. 16 (a). Their internal rate of return (IRR) and return on investment (ROI) are 12.60%/78.97%, 12.17%/68.31%, 8.32%/32.45%. as shown in Fig. 16 (b). The LCOE (levelized cost of electricity) of W190–P10-E95-B30, W120–P80-E80-B25, W60–P140-E85-B27.5 are 0.2692¥/kWh, 0.2886¥/kWh, 0.3617¥/kWh as shown in Fig. 16 (c). The global weight average LCOE of onshore wind and solar PV in 2021 (the data for 2022 is not published yet) are 0.2283¥/kWh, 0.3332¥/kWh [65]. Their LCOEs order from low to high as global average onshore wind < W190-PV10-E95-B30<W120–P80-E80-B25< global average solar PV < W60–P140-E85-B27.5. Increasing the wind-to-generation capacity ratio can reduce the LCOE of the WPEB system due to the higher capacity factor. The LCOH (using alkaline electrolyzers) of W190–P10-E95-B30, W120–P80-E80-B25, W60–P140-E85-B27.5 at 2022 are 10.6248¥/kg, 32.8202¥/kg, 43.4450¥/kg as shown in Fig. 16 (d). The Department for Business, Energy and Industrial Strategy (BEIS) estimates the cost of using curtailed electricity for hydrogen production and reports that the LCOH of alkaline electrolyzers in 2020, 2025, 2035, 2040, 2045, 2050 are 15.5051¥/kg, 13.9451¥/kg, 12.6460¥/kg, 12.3646¥/kg, 12.2164¥/kg, 12.0690¥/kg (calculated using the exchange rate of 1GBP = 8.3CNY). The case of W190–P10-E95-B30 is cost-competitive in both LCOE and LCOH as shown in Fig. 16 (c) and Fig. 16 (d).

The net present cost (NPC) of W190–P10-E95-B30, W120–P80-E80-B25, W60–P140-E85-B27.5 are 2675.73 million¥, 2350.38 million¥, 2114.53 million¥. Take W190–P10-E95-B30 as an example, CAPEX accounts for 73.30% of the total NPC namely the fixed cost makes up the vast majority as shown in Fig. 16 (e). The electrolysis station is an expensive component of the WPEB system which accounts for 44.65%, 42.81%, 50.56% of the NPC for W190–P10-E95-B30, W120–P80-E80-B25, W60–P140-E85-B27.5 as shown in Fig. 16 (f). The unit CAPEX of the wind farm (5917¥/kW) is greater than the PV farm (4633¥/kW) as shown in Table 4. However, the higher capacity factor of wind farm results in a lower LCOE compared to PV farm as shown in Fig. 16 (c).

3.3. *Research implication, limitation and prospect*

During the planning & design stage of renewable energy-based hydrogen production, the capacity configuration of the renewable power source and electrolysis station is the top priority. The system capacity configuration influences the component capacity factor, excess electricity rate (especially for off-grid systems), CAPEX and finally the NPC, IRR. This work provides a WPEB system capacity optimization method by combining the 8760-h production simulation and gradient

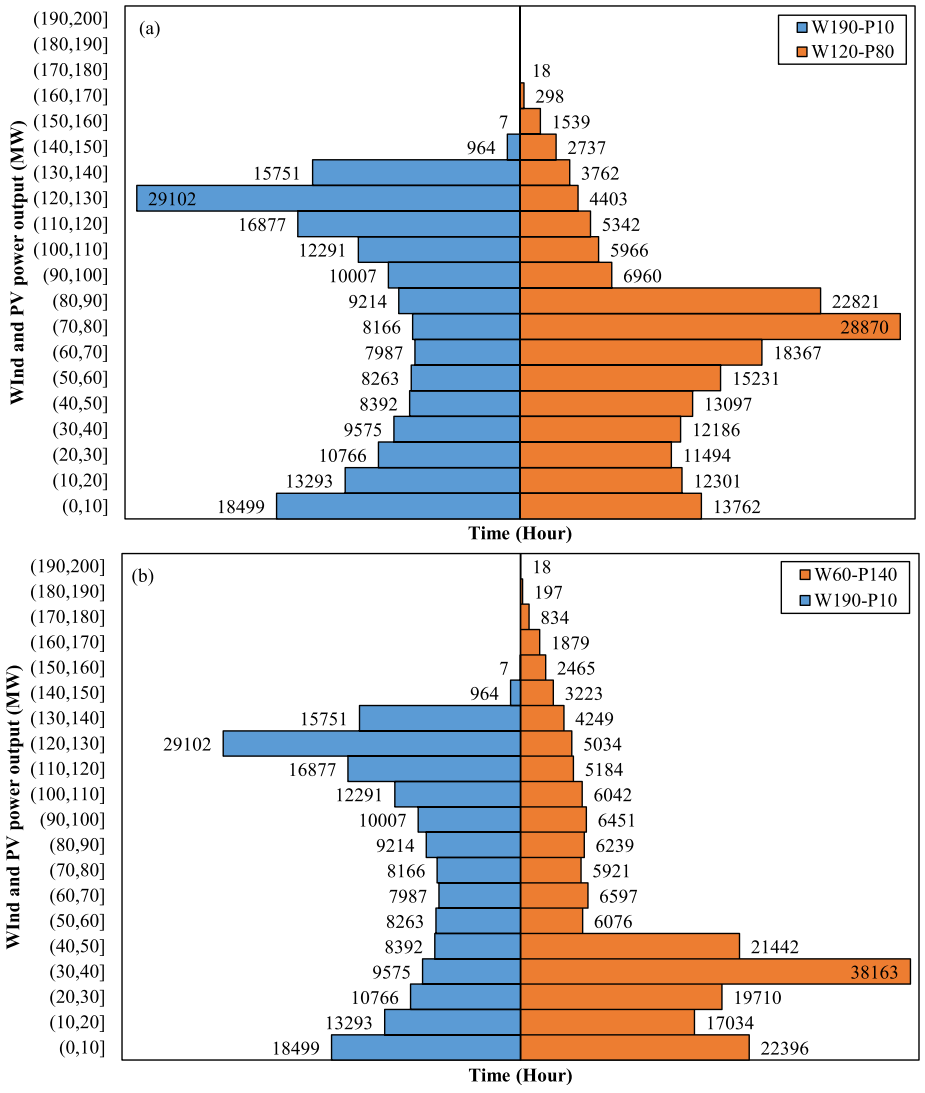


Fig. 13 – Frequency distribution histogram of wind-PV power output using 2001–2021 weather patterns.

descent method. The iteration of the component capacity is based on the typical meteorological year and 8760-h production simulation which improves the system's economic performance in real-world operation.

The limitation and prospect of this work is summarized below: (1) This work takes LCOE as the single objective function for capacity optimization. A multi-objective optimization program that including NPC (or LCOE, LCOH), excess electricity rate, etc. should be considered in future research. (2) The wind power and solar power installed capacity is constant at 200 MW of this work. The total installed capacity of wind power and solar power can be varied and its impact on the excess electricity rate can be discussed. (3) This work assumed that the hydrogen load is high enough, namely all the produced hydrogen can be consumed downstream. The impact of hydrogen load profile volatility and uncertainty on the capacity configuration can be studied in future research. Similarly, the impact of the electric load profile for the grid on the optimization result can be studied even though there is no requirement in the real-world operation currently. (4) The battery bank power capacity equals to or slightly greater than the minimum load ratio of the electrolysis station (30% in this work) and the discharge duration is 1 h of this work. The battery bank power capacity can become a decision variable and the effect of discharge duration on the system cost, excess electricity rate, electrolysis capacity factor are the interesting topics in the further study.

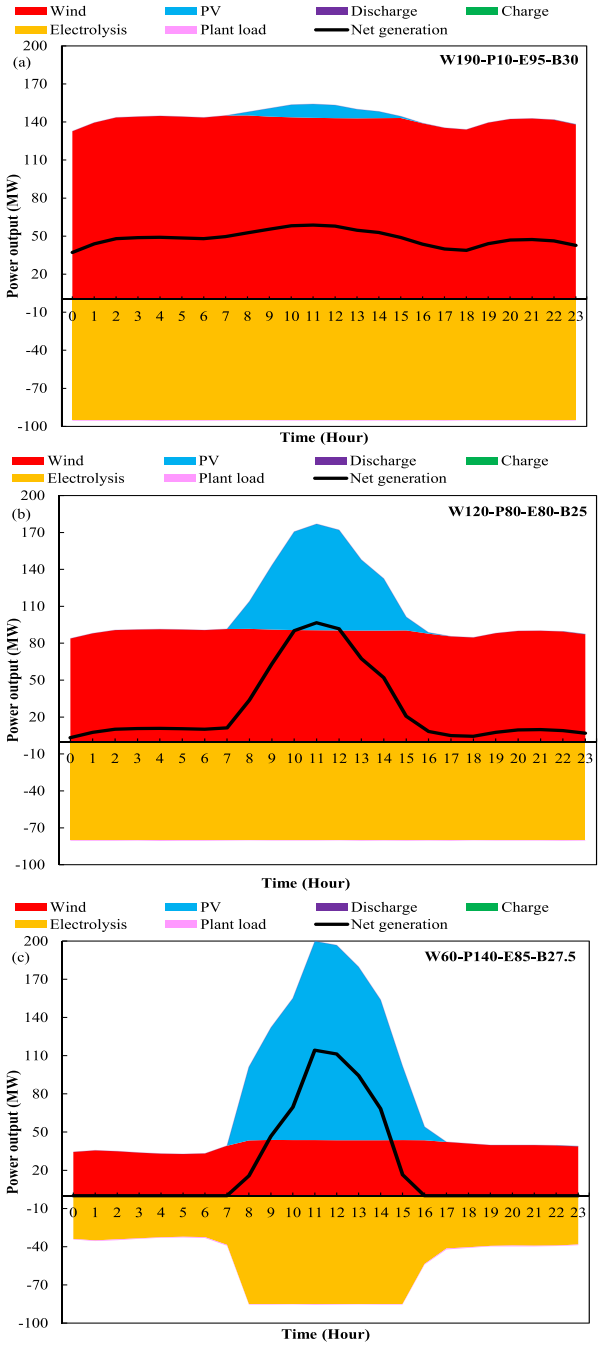


Fig. 14 – Typical daily operation of maximum wind-PV power for (a) W190-PV10-E95-B30, using 2021-01-06 weather pattern, (b) W120-PV80-E80-B25, using 2021-01-06 weather pattern, (c) W60-PV140-E85-B27.5, using 2012-02-06 weather pattern.

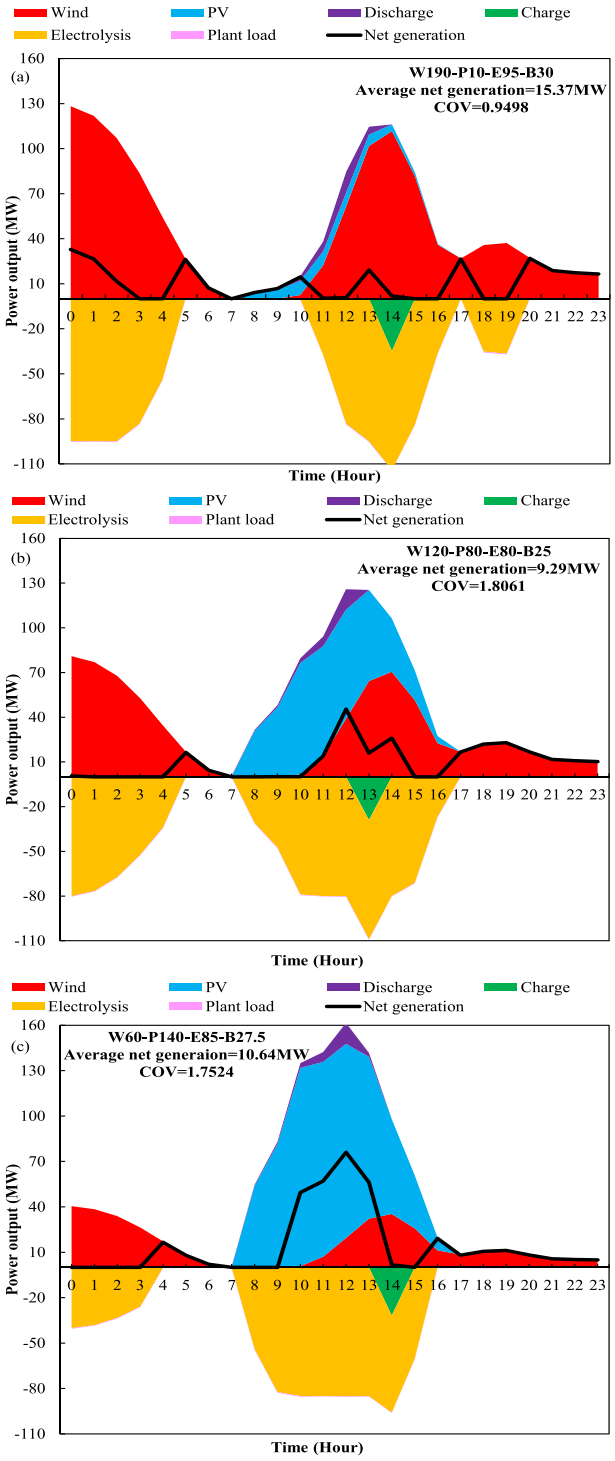


Fig. 15 – Typical daily operation of fluctuated net generation for (a) W190-PV10-E95-B30, (b) W120-PV80-E80-B25, (c) W60-PV140-E85-B27.5 using 2021-02-13 weather pattern.

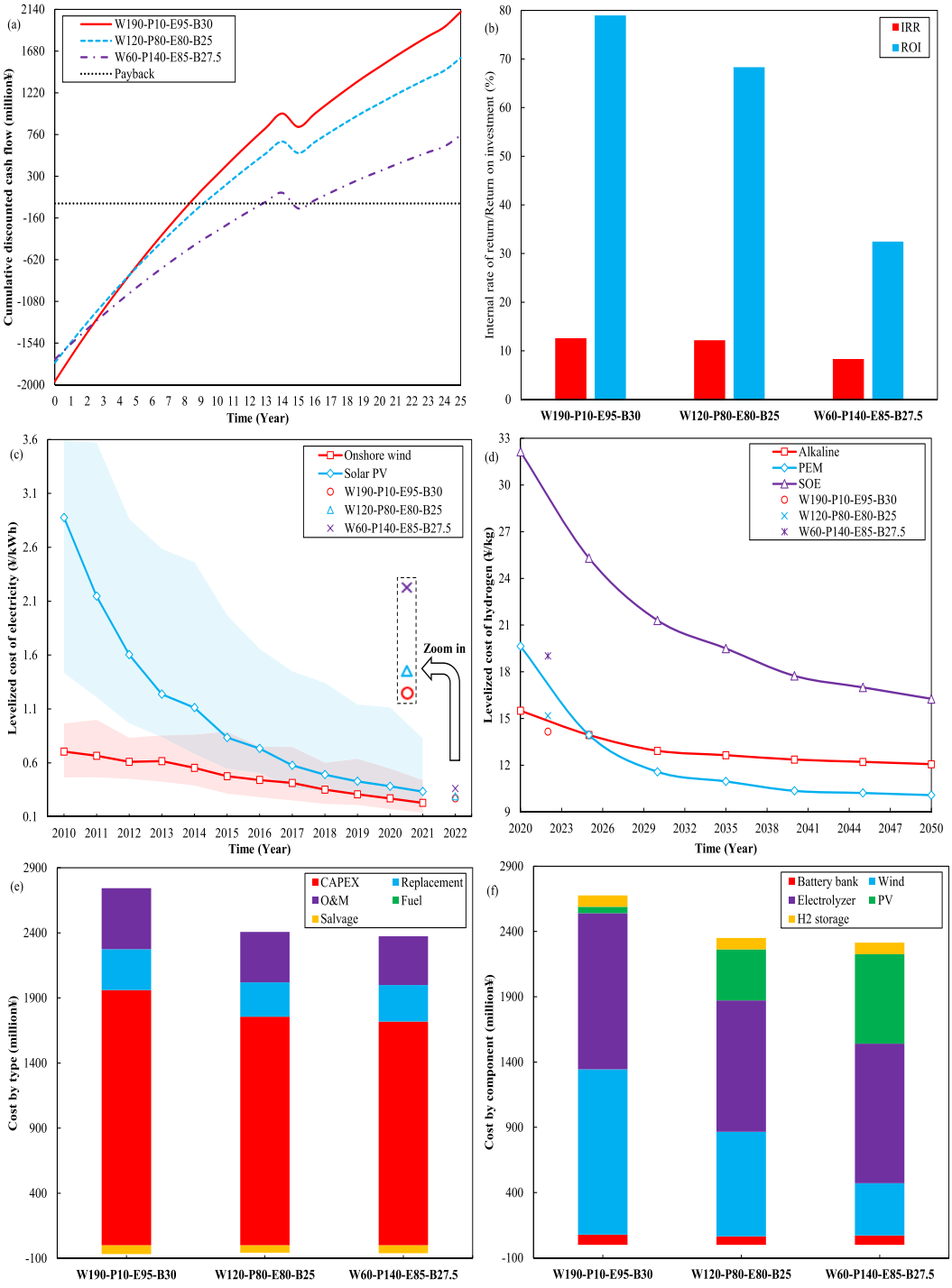


Fig. 16 – (a) Cash flow and net present value; (b) internal rate of return, return on investment; (c) levelized cost of electricity between this work and International Renewable Energy Agency (IRENA)data [65]; (d) levelized cost of hydrogen between this work and Department for Business, Energy and Industrial Strategy (BEIS) data [68]; (e) cost by type; (f) cost by component.

4. Conclusions

Utility-scale Wind-Photovoltaic-Electrolysis-Battery (WPEB) system is an emerging technology for large-scale hydrogen production and curtailment reduction in the area with abundant wind & solar energy resources. Capacity configuration is the first step of the power system planning & design which determines the system life cycle technoeconomic performance. This work proposes a WPEB system capacity optimization method that applies to those adopting an open loop "Power-to-H_2" structure for hydrogen production. The system in Damao Banner (total wind-PV power installed capacity 200 MW) is taken as a case study. The results are summarized below.

1. The levelized cost of electricity (LCOE) is used as the objective function and the wind-to-generation capacity ratio and electrolyzer-to-generation capacity ratio are used as decision variables. The constraints contain power balance, annual grid sales rate, wind/PV/electrolysis station capacity constraints. The feasible domain and the optimized solution are figured out by combining the 8760-h production simulation and gradient descent method. The optimized capacity configuration is 190 MW wind farm-10 MW PV farm-95MW Electrolysis station-30MW/30 MWh battery bank (W190–P10-E95-B30) at the studied site.
2. Increasing the wind-to-generation capacity ratio can reduce the LCOE, required grid sales capacity, net generation curve fluctuation. The electrolysis station capacity is co-determined by the annual grid sales rate constraint and LCOE objective. The battery bank time-shifts electricity to reduce the electrolysis station downtime. Thus, its power capacity should be equal to or slightly higher (depending on the battery pack capacity) than the minimum load ratio of the electrolysis station. The discharge duration should be no less than 1 h.
3. The net present value (NPV), net present cost (NPC), internal rate of return (IRR), return on investment (ROI), levelized cost of electricity (LCOE), levelized cost of hydrogen (LCOH) and payback period of W190–P10-E95-B30 are 2113.08 million¥, 2675.73 million¥, 12.60%, 78.97%, 0.2692¥/kWh, 14.1574¥/kg and 9 years.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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